

2.9 Constrained Linear Models

In Models I—VII, $\underline{\beta}$ can assume any value in the Euclidean p -space. We now consider the case where $\underline{\beta}$ is restricted to the subspace of R^p consisting of $\underline{\beta}$ values that satisfy

$$A\underline{\beta} = \underline{b},$$

where A is a given $q \times p$ matrix of rank $q^* \geq 1$ and $\underline{b}$ is a given $q \times 1$ vector. To avoid trivialities, we assume that the linear system $A\underline{\beta} = \underline{b}$ is consistent. We will refer to those modified versions of Models I—VII in which $\underline{\beta}$ is restricted to this subspace as Models Ia—VIIa.

Definition. Under Models IIa—VIIa, an estimator $t(\underline{y})$ is said to be an unbiased estimator of a linear function $\underline{\lambda}'\underline{\beta}$ if $E[t(\underline{y})] = \underline{\lambda}'\underline{\beta}$ for all $\underline{\beta}$ satisfying $A\underline{\beta} = \underline{b}$.

Lemma 2.9. *Under Models IIa—VIIa, a linear estimator $c + \underline{a}'\underline{y}$ of a linear function $\underline{\lambda}'\underline{\beta}$ is unbiased if and only if there exists a vector $\underline{k}$ such that*

$$c = \underline{k}'\underline{b} \quad \text{and} \quad \underline{\lambda}' = \underline{a}'X + \underline{k}'A.$$

Proof. $E(c + \underline{a}'\underline{y}) = \underline{\lambda}'\underline{\beta}$ for all $\underline{\beta}$ satisfying $A\underline{\beta} = \underline{b}$

$$\iff c + \underline{a}'X\underline{\beta} = \underline{\lambda}'\underline{\beta} \text{ for all } \underline{\beta} \text{ satisfying } A\underline{\beta} = \underline{b}$$

$$\iff c + (\underline{a}'X - \underline{\lambda}')\underline{\beta} = 0 \text{ for all } \underline{\beta} \text{ satisfying } A\underline{\beta} = \underline{b}$$

It remains to show that if $c = \underline{k}'\underline{b}$ and $\underline{\lambda}' = \underline{a}'X + \underline{k}'A$ for some $\underline{k}$, then

$$c = (\underline{\lambda}' - \underline{a}'X)A^-\underline{b} \text{ and } \underline{\lambda}' = \underline{a}'X + (\underline{\lambda}' - \underline{a}'X)A^-A.$$

Definition. Under Models IIa—VIIa, a linear function $\underline{\lambda}'\underline{\beta}$ is said to be estimable if there exists a linear unbiased estimator of $\underline{\lambda}'\underline{\beta}$, that is, if there exists a linear estimator $c + \underline{a}'\underline{y}$ such that $E(c + \underline{a}'\underline{y}) = \underline{\lambda}'\underline{\beta}$ for all $\underline{\beta}$ satisfying $A\underline{\beta} = \underline{b}$.

Theorem 2.11. *Under Models IIa—VIIa, a linear function $\underline{\lambda}'\underline{\beta}$ is estimable if and only if $\underline{\lambda}' \in \mathcal{R} \begin{bmatrix} X \\ A \end{bmatrix}$.*

We now consider the least squares estimation of an estimable function $\underline{\lambda}'\underline{\beta}$ under Models IIa—VIIa.

Definition. Under Models IIa—VIIa, a least squares estimate of an estimable function $\underline{\lambda}'\underline{\beta}$ is $\underline{\lambda}'\hat{\underline{\beta}}$, where $\hat{\underline{\beta}}$ is any value of $\underline{\beta}$ that minimizes $Q(\underline{\beta}) \equiv (\underline{y} - X\underline{\beta})'(\underline{y} - X\underline{\beta})$ for $\underline{\beta}$ satisfying $A\underline{\beta} = \underline{b}$.

The Lagrangian function for the problem of minimizing $Q(\underline{\beta})$ with respect to $\underline{\beta}$, subject to the constraints $A\underline{\beta} = \underline{b}$, is

$$L(\underline{\beta}, \underline{\rho}) = Q(\underline{\beta}) + 2\underline{\rho}'(A\underline{\beta} - \underline{b}).$$

We have

$$\frac{\partial L}{\partial \underline{\beta}} = -2X'\underline{y} + 2X'X\underline{\beta} + 2A'\underline{\rho} \quad \text{and} \quad \frac{\partial L}{\partial \underline{\rho}} = 2(A\underline{\beta} - \underline{b}).$$

Equating these partial derivatives to zero gives

$$X'X\underline{\beta} + A'\underline{\rho} = X'\underline{y} \quad \text{and} \quad A\underline{\beta} = \underline{b}.$$

Thus, a necessary condition for $Q(\underline{\beta})$ to assume a minimum at $\underline{\beta} = \hat{\underline{\beta}}$, for $\underline{\beta}$ satisfying $A\underline{\beta} = \underline{b}$, is that the equations

$$\begin{bmatrix} X'X & A' \\ A & 0 \end{bmatrix} \begin{bmatrix} \hat{\underline{\beta}} \\ \underline{\rho} \end{bmatrix} = \begin{bmatrix} X'\underline{y} \\ \underline{b} \end{bmatrix}$$

have a solution whose “first” component is $\hat{\underline{\beta}}$. We refer to these equations as the constrained normal equations.

The following lemma can be used to establish the consistency of the constrained normal equations.

Lemma 2.10. *The matrix equation $BW = D$ is consistent, that is, has a solution for W , if and only if $\mathcal{C}(D) \subset \mathcal{C}(B)$ if and only if $\mathcal{C}^\perp(B) \subset \mathcal{C}^\perp(D)$ that is, if and only if $\underline{a}'D = \underline{0}'$ for every vector $\underline{a}$ such that $\underline{a}'B = \underline{0}'$.*

Theorem 2.12. *The constrained normal equations are consistent.*

Proof. Take $\underline{a} = \begin{bmatrix} \underline{a}_1 \\ \underline{a}_2 \end{bmatrix}$ to be any vector such that

$$\underline{a}' \begin{bmatrix} X'X & A' \\ A & 0 \end{bmatrix} = (\underline{a}_1', \underline{a}_2') \begin{bmatrix} X'X & A' \\ A & 0 \end{bmatrix} = (\underline{a}_1'X'X + \underline{a}_2'A, \underline{a}_1'A') = \underline{0}'.$$

By Lemma 2.10, it suffices to show that

$$\underline{a}' \begin{bmatrix} X'y \\ \underline{b} \end{bmatrix} = (\underline{a}_1', \underline{a}_2') \begin{bmatrix} X'y \\ \underline{b} \end{bmatrix} = \underline{a}_1'X'y + \underline{a}_2'\underline{b} = 0.$$

Theorem 2.13. $Q(\underline{\beta})$ assumes a minimum at $\underline{\beta} = \widehat{\underline{\beta}}$, for $\underline{\beta}$ satisfying $A\underline{\beta} = \underline{b}$, if and only if the constrained normal equations have a solution whose first component is $\widehat{\underline{\beta}}$.

Proof. Let $\begin{bmatrix} \widehat{\underline{\beta}} \\ \underline{\rho} \end{bmatrix}$ represent any solution to the constrained normal equations, and let $\widetilde{\underline{\beta}}$ represent any solution to $A\underline{\beta} = \underline{b}$.

Then, $X'X\widehat{\underline{\beta}} + A'\underline{\rho} = X'y$, $A\widehat{\underline{\beta}} = \underline{b}$, and $A\widetilde{\underline{\beta}} = \underline{b}$. We have

$$\begin{aligned} Q(\widetilde{\underline{\beta}}) &= (\underline{y} - X\widetilde{\underline{\beta}})'(\underline{y} - X\widetilde{\underline{\beta}}) = [(\underline{y} - X\widehat{\underline{\beta}}) + X(\widehat{\underline{\beta}} - \widetilde{\underline{\beta}})]'[(\underline{y} - X\widehat{\underline{\beta}}) + X(\widehat{\underline{\beta}} - \widetilde{\underline{\beta}})] \\ &= Q(\widehat{\underline{\beta}}) + [X(\widehat{\underline{\beta}} - \widetilde{\underline{\beta}})]'[X(\widehat{\underline{\beta}} - \widetilde{\underline{\beta}})] + 2[X(\widehat{\underline{\beta}} - \widetilde{\underline{\beta}})]'(\underline{y} - X\widehat{\underline{\beta}}). \end{aligned}$$

Furthermore, if $\underline{\beta}^*$ is any other value of $\underline{\beta}$ at which $Q(\underline{\beta})$ assumes a minimum, for $\underline{\beta}$ satisfying $A\underline{\beta} = \underline{b}$, then $Q(\underline{\beta}^*) = Q(\widehat{\underline{\beta}})$. Thus, $[X(\widehat{\underline{\beta}} - \underline{\beta}^*)]'[X(\widehat{\underline{\beta}} - \underline{\beta}^*)] = 0$. This implies that $X(\widehat{\underline{\beta}} - \underline{\beta}^*) = \underline{0}$, or $X\underline{\beta}^* = X\widehat{\underline{\beta}}$. Then,

$$X'X\underline{\beta}^* + A'\underline{\rho} = X'X\widehat{\underline{\beta}} + A'\underline{\rho} = X'y,$$

so that $\begin{bmatrix} \underline{\beta}^* \\ \underline{\rho} \end{bmatrix}$ is also a solution to the constrained normal equations.

We now establish some properties of the coefficient matrix of the constrained normal equations.

Lemma 2.11.

$$\text{rank} \begin{bmatrix} X'X & A' \\ A & 0 \end{bmatrix} = \text{rank} \begin{bmatrix} X'X \\ A \end{bmatrix} + \text{rank}(A).$$

Proof. It suffices to show that $\mathcal{C} \begin{bmatrix} X'X \\ A \end{bmatrix}$ and $\mathcal{C} \begin{bmatrix} A' \\ 0 \end{bmatrix}$ are essentially disjoint, that is, have only the zero vector in common, because this implies that the union of any set of linearly independent columns of $\begin{bmatrix} X'X \\ A \end{bmatrix}$ and any set of linearly independent columns of $\begin{bmatrix} A' \\ 0 \end{bmatrix}$ is itself a set of linearly independent column vectors.

Lemma 2.12. (i) $\mathcal{R} \begin{bmatrix} X'X \\ A \end{bmatrix}$ matches $\mathcal{R} \begin{bmatrix} X \\ A \end{bmatrix}$. (ii) $\text{rank} \begin{bmatrix} X'X \\ A \end{bmatrix} = \text{rank} \begin{bmatrix} X \\ A \end{bmatrix}$.

Corollary to Lemma 2.11.

$$\text{rank} \begin{bmatrix} X'X & A' \\ A & 0 \end{bmatrix} = \text{rank} \begin{bmatrix} X \\ A \end{bmatrix} + \text{rank}(A).$$

Theorem 2.14. Let $G = \begin{bmatrix} G_{11} & G_{12} \\ G_{21} & G_{22} \end{bmatrix}$ represent any generalized inverse of $\begin{bmatrix} X'X & A' \\ A & 0 \end{bmatrix}$.
Then,

(i) $A'G'_{12}A' = A'G_{21}A' = A'$;

(ii) $-A'G_{22}A = -A'G'_{22}A = X'XG'_{21}A = A'G_{21}X'X = X'XG_{12}A = A'G'_{12}X'X$;

(iii) $AG_{11}A'$, $AG_{11}X'X$, and $X'XG_{11}A'$ are all zero matrices;

(iv) $X'XG_{11}X'XG_{11}X'X = X'XG_{11}X'X = X'XG'_{11}X'XG_{11}X'X = X'XG'_{11}X'X$;

(v) $\text{tr}(X'XG_{11}) = \text{rank} \begin{bmatrix} X'X \\ A \end{bmatrix} - \text{rank}(A) = \text{tr}(X'XG'_{11})$;

(vi) $X'XG_{11}X'X$ and $A'G_{22}A$ are invariant for any choice of G_{11} and G_{22} .

Lemma 2.13. *For any generalized inverse*

$$G = \begin{bmatrix} G_{11} & G_{12} \\ G_{21} & G_{22} \end{bmatrix} \quad \text{of} \quad \begin{bmatrix} X'X & A' \\ A & 0 \end{bmatrix},$$

$$X'X = X'XG_{11}X'X - A'G_{22}A.$$

Theorem 2.15. *Under Models IIa—VIIa, $\underline{\lambda}'\widehat{\underline{\beta}} = \underline{\lambda}'\widetilde{\underline{\beta}}$ for the first components $\widehat{\underline{\beta}}$ and $\widetilde{\underline{\beta}}$ of any two solutions to the constrained normal equations if and only if $\underline{\lambda}'\underline{\beta}$ is estimable.*

Proof. Let $G = \begin{bmatrix} G_{11} & G_{12} \\ G_{21} & G_{22} \end{bmatrix}$ represent any generalized inverse of $\begin{bmatrix} X'X & A' \\ A & 0 \end{bmatrix}$.

By Theorem 1.3, all solutions to the constrained normal equations are

$$\begin{aligned} & \begin{bmatrix} G_{11} & G_{12} \\ G_{21} & G_{22} \end{bmatrix} \begin{bmatrix} X'\underline{y} \\ \underline{b} \end{bmatrix} + \left\{ \begin{bmatrix} I & 0 \\ 0 & I \end{bmatrix} - \begin{bmatrix} G_{11} & G_{12} \\ G_{21} & G_{22} \end{bmatrix} \begin{bmatrix} X'X & A' \\ A & 0 \end{bmatrix} \right\} \underline{z} \\ &= \begin{bmatrix} G_{11}X'\underline{y} + G_{12}\underline{b} \\ G_{21}X'\underline{y} + G_{22}\underline{b} \end{bmatrix} + \begin{bmatrix} I - G_{11}X'X + G_{12}A & -G_{11}A' \\ -G_{21}X'X + G_{22}A & I - G_{21}A' \end{bmatrix} \underline{z} \end{aligned}$$

as $\underline{z}$ ranges over the space of all possible vectors. Thus, $\underline{\lambda}'\widehat{\underline{\beta}} = \underline{\lambda}'\widetilde{\underline{\beta}}$ for any two solutions to the constrained normal equations if and only if

$$\underline{\lambda}'[I - G_{11}X'X - G_{12}A, -G_{11}A']\underline{z} = 0 \quad \text{for all } \underline{z}$$